

Spring 25 Q1 (a)

$$1 \mu s = 1000 \text{ ns}$$

given,

CPU $\longrightarrow$ device: 0.2 M: 200 ns

device $\rightarrow$ cpu : $0.4 \mu\text{s} = 400 \text{ ns}$

rate: 96 kb/s

8 Byte transferred at once.

total size = ? , Assume n Byte

N.B চিহ্নে mainly total টা যেই ব্যয়-ভুক্ত ব্যবস্থা transfer হইছে তা (যদি ~~সব ব্যয় ও বি~~ ~~সমস্ত ব্যয়~~ কালেক্ট হবে।) ($\frac{24}{8}$)

time for one cycle

number of cycles

$$t = (t_1 + t_2 + t_3) \times n$$

$$\Rightarrow t = (t_1 + t_2 + t_3) \times \frac{2L}{S}$$

$$\Rightarrow 62960800^{ns} = \left(200^{ns} + \frac{1 \text{ KB}}{96 \text{ KB/s}} + 400^{ns} \right) \times \frac{1}{8}$$

$$\Rightarrow (62960800 \times 8) \text{ ns} = 16200 \text{ ns} + \frac{1}{96 \times 2^{10}} \times 10^9 \text{ ns} + 400 \mu$$

$$\Rightarrow 503686400 = \frac{5 \times 10^9}{96 \times 10^3} + 600n$$

$$\Rightarrow \frac{10^9}{26 \times 10^6} u + 600u - 503686400 = 0$$

$$\Rightarrow n = 222.9887975 \text{ B}_{\text{He}}$$

Assignment math 84

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CPU $\rightarrow$ device : 150 ns

device $\rightarrow$ CPU : 550 ns

total data - : 2048 Byte

total time : 13335083 ns

rate of

$$t = t_1 + t_2 + t_3$$

$$\Rightarrow 133350853 = 150 + \frac{2048 \times 10^7}{n} + 550$$

$$\Rightarrow n = 153587.91 \text{ Bytes}$$

given,

$$t_1 = 150 \text{ ns}, t_3 = 550 \text{ ns}$$

total = 7875 Byte, rate = 153587.91 B/s

Main সহায় হলে চিহ্নে current সহ, আদার
সহায় (অথ Data সহ সহ) (বাকি মাধ্যম) 50,
total সহায় মাধ্যমে বা সহ সহায় সহ সহায়
সহ সহায় (Direct- $\frac{\text{total}}{\text{rate}}$) সহ সহ সহ সহ সহ

সেং-সং?

$$1+2+3+\dots+n = 7875$$

$$\Rightarrow \frac{n(n+1)}{2} = 7875$$

$$\Rightarrow n(n+1) - (7875 \times 2) = 0$$

$$\Rightarrow n = 125, \text{ (126) } \rightarrow \text{times -ve 24 না।}$$

কিন্তু total time এর একটি প্রতিসার আলম
আলম time লাগবে, যার file size same না।

$$t = (t_1 + t_2 + t_3) + (t_1 + t_2 + t_3) + (t_1 + t_2 + t_3) + \dots + (t_1 + t_2 + t_3)$$

1B 2B 3B 125B

$$\Rightarrow t = 125t_1 + 125t_3 + (t_2 + t_2 + t_2 + \dots + t_2)$$

$$\Rightarrow t = 125(150 + 550) + \left(\frac{1 \times 10^9}{n} + \frac{2 \times 10^9}{n} + \dots + \frac{125 \times 10^9}{n} \right)$$

$$\Rightarrow t = 125 \times 700 + \left[\frac{10^9}{n} (1+2+3+\dots+125) \right]$$

$$\Rightarrow t = 125 \times 700 + \left(\frac{10^9}{n} \times 7875 \right)$$

(1+2+...+n = 7875)

$$\Rightarrow t = 51361067.09 \text{ ns} \rightarrow 153587.91 \text{ B/s}$$

Fall 25 81(a)

$$t_1 = 600 \text{ ns}$$

$$t_3 = 500 \text{ ns}$$

rate : 40 MB/s

total : 600 byte

even cycle 5 5 Byte send ~~2~~ } Total cycle
 Odd " " 3 " " " Ber ~~2~~ hint

Total n ~~2~~

$$\frac{3+5}{1} + \frac{3+5}{2} + \dots + = 600$$

$$n(3+5) = 600$$

$$n = 75$$

3 Bytes sent 75 times
 5 " " 75 "
 total cycle = 150 times.

$$t = \overset{\rightarrow 3 \text{ Bytes}}{(t_1 + t_2 + t_3)} 75 + \overset{\rightarrow 5 \text{ Bytes}}{(t_1 + t_2 + t_3)} 75$$

$$= 75 \left(600 + \frac{3 \times 10^9}{40 \times 2^{10}} + 500 \right) + 75 \left(600 + \frac{5 \times 10^9}{40 \times 2^{10}} + 500 \right)$$

$$= 14813437.5 \text{ ns}$$